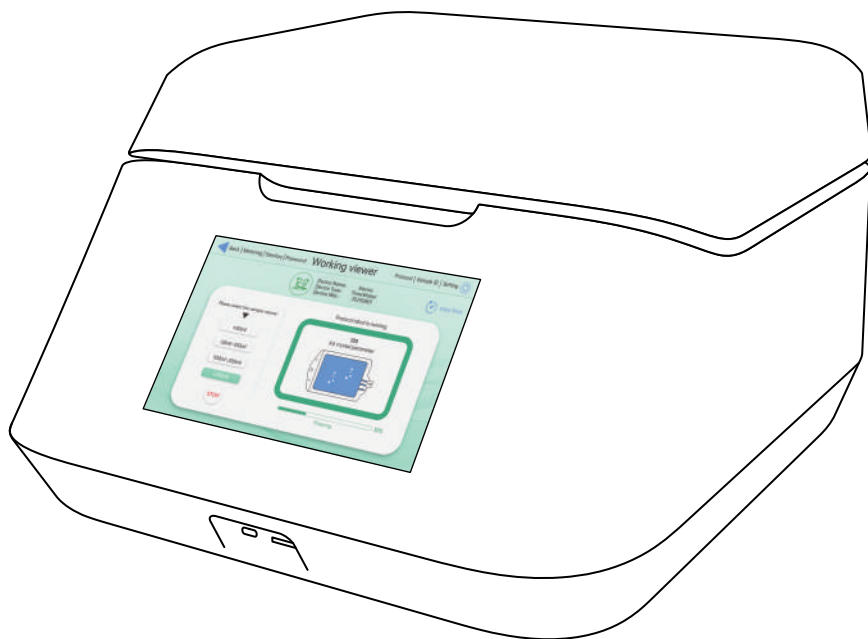


ThawWaker



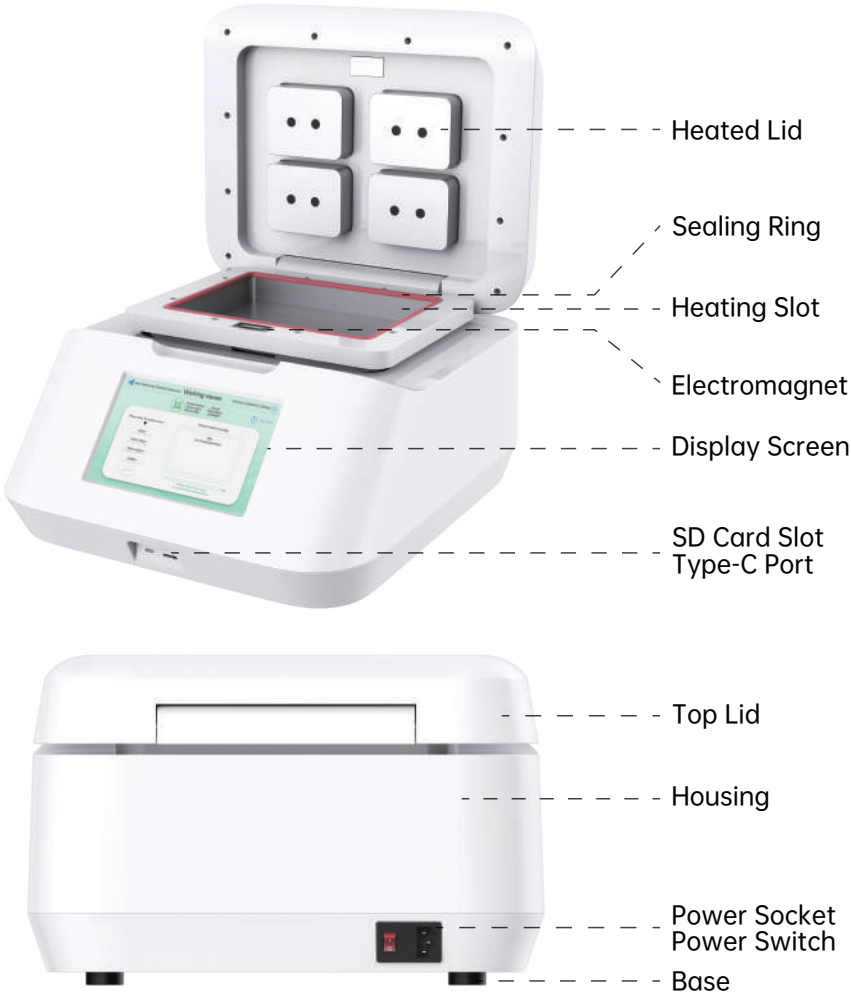
User Manual

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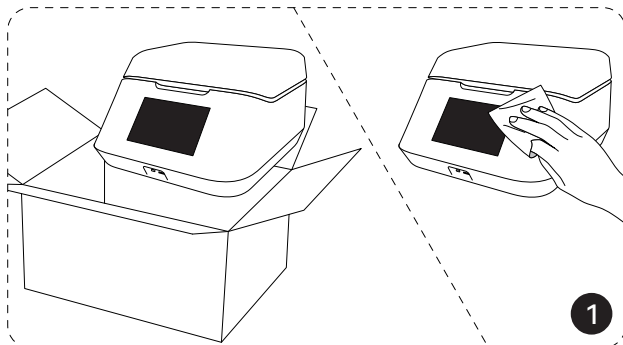
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Function description and appearance

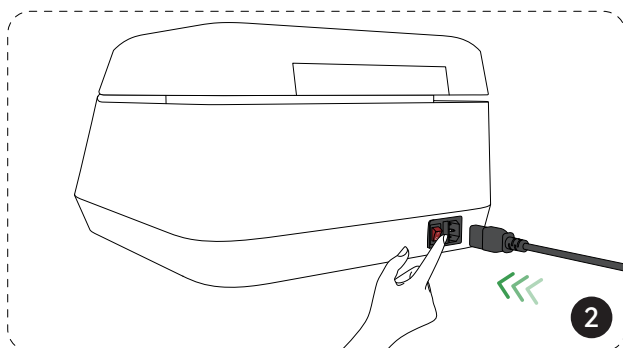
The freezing, thawing, and cryopreservation of samples are receiving increasing attention from researchers. As essential steps in biological experiments, their stability and contamination rate directly affect experimental results. Therefore, high-quality equipment can better assist researchers in conducting experiments and obtaining results more efficiently and reliably. Similarly, industrial users also require such products to ensure stable production and improve efficiency. Kemesser has focused on the research and development of such technologies for over ten years. Its product series effectively supports the processes of biological sample freezing, thawing, and cryopreservation, providing customers with a more reliable and stable operating environment. The cell thawer is designed for efficient thawing of viable cells stored in containers of various specifications. After biological containers are taken directly from liquid nitrogen or dry ice, they are placed into the slot of the cell thawer. The thawing system uses sensor technology and temperature control technology to rapidly restore cellular biological activity, achieving effective cell recovery.



Quick Start Guide



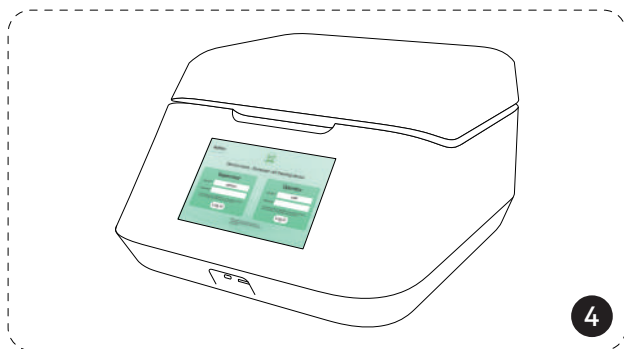
Remove the device from the carton and wipe its surface.



Carefully connect the power cord from the accessories to the rear power socket and turn on the power switch.

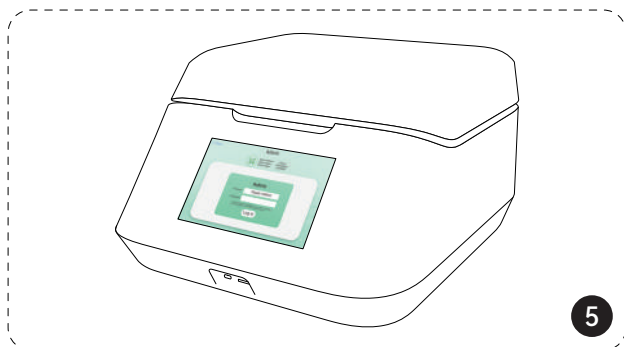


Enter the welcome screen and wait for the countdown at the top right to finish before proceeding to the login screen.



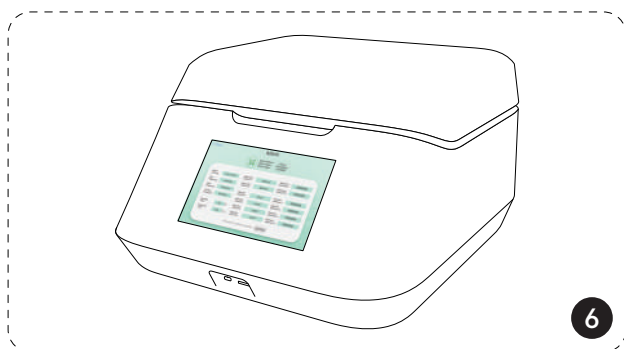
Click **Admin** at the top left of the login screen.

4



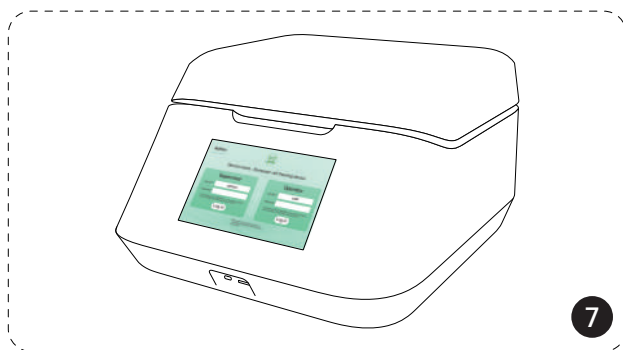
Enter the password as an admin.

5



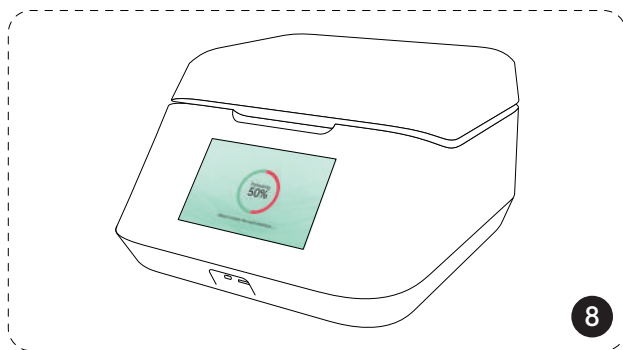
Enter the **Admin** interface to assign accounts and set passwords. After completing the settings, click Confirm to save the data.

6



Enter the password as a supervisor.

7



Wait for preheating.

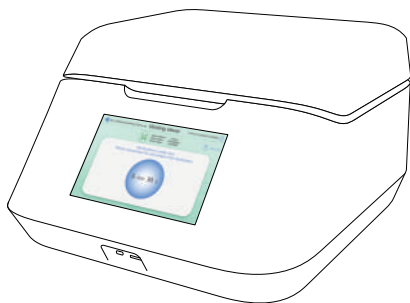
8



After preheating is complete and the work interface appears, click the top right corner to enter Settings, adjust the ice crystal and other parameters, then click Confirm to save the data.

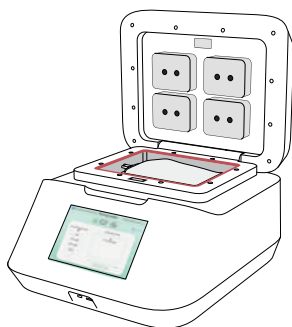
9

Note: Ice crystal parameters are a critical factor in the thawing process. Since each user's sample volume and cryopreservation solution may vary, we strongly recommend performing trial thawing with the same type and volume of cryopreservation solution before actual use to determine the optimal ice crystal settings. The ideal thawing state is achieved when 5%-15% of ice crystals remain at the end of the process.



Return to the work interface, click **【Sterilize】** at the top left to enter the sterilization screen, and wait for the process to complete.

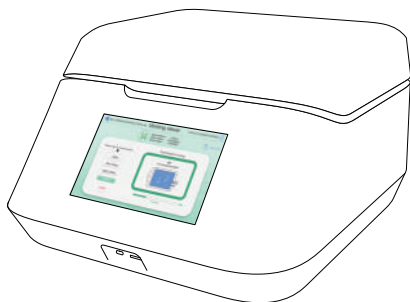
10



Return to the main interface, select the appropriate cryobag volume, choose the corresponding volume on the device interface, open the top lid, place the cryobag into the sample slot, and close the lid to start cell thawing.

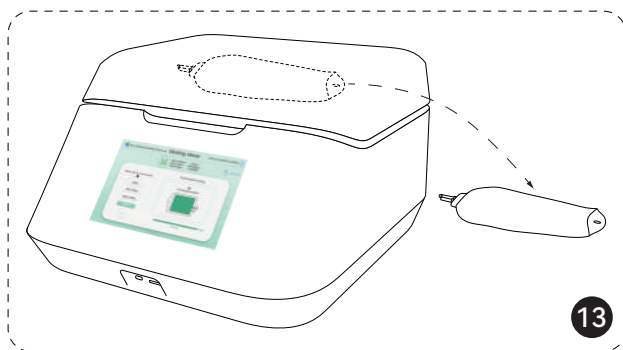
11

Note: If the sample in the bag is unevenly distributed, ensure that the area with the largest volume is aligned with the black reference point on the device lid.

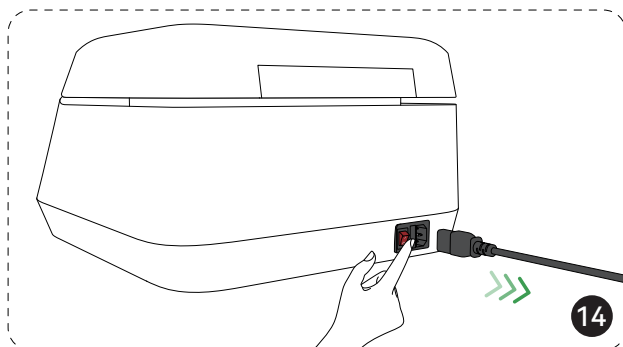


Enter the thawing interface, cells are currently being thawed.

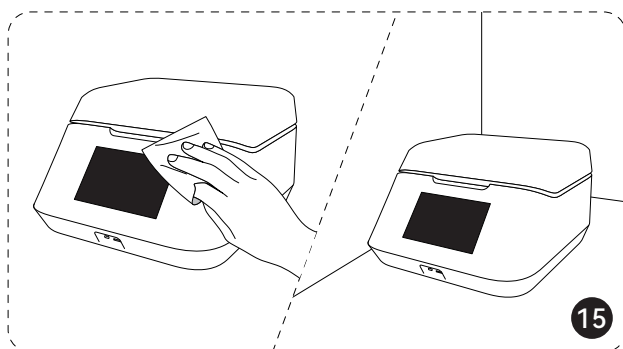
12



After cell thawing is complete, promptly remove the cryobag samples.

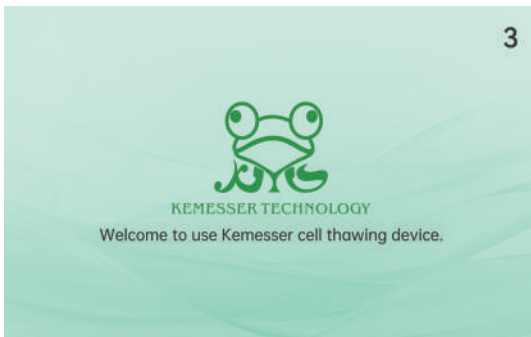
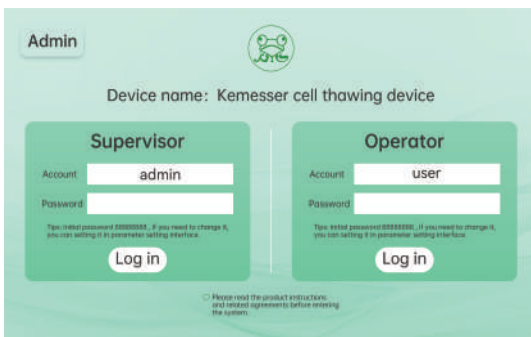
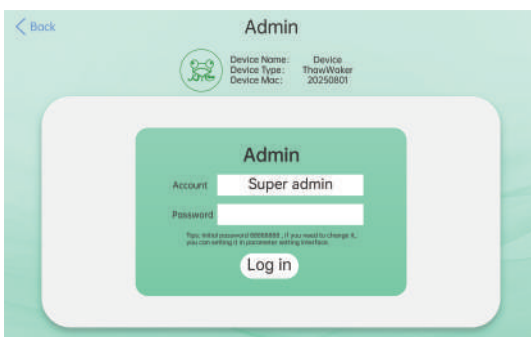


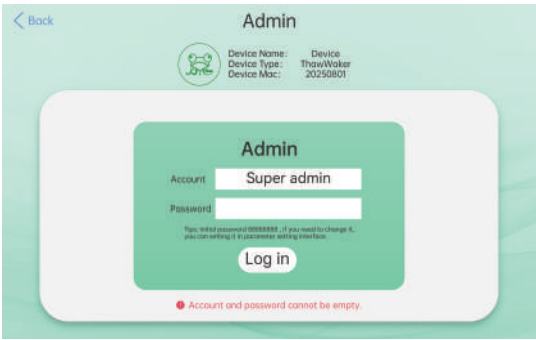
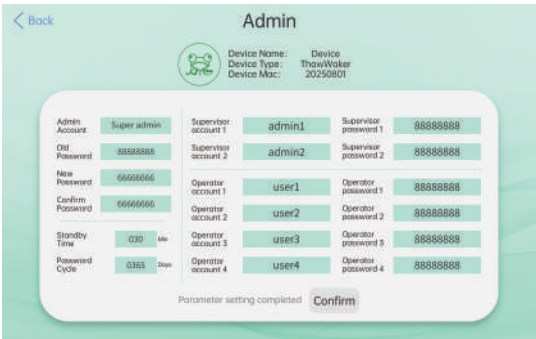
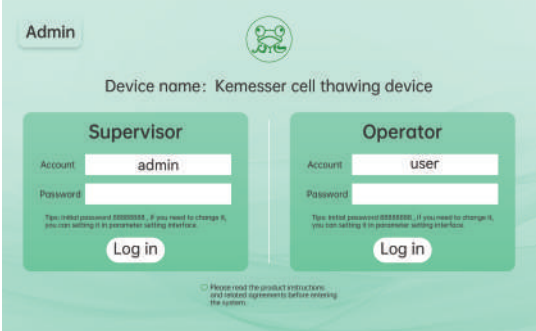
After placing the cryobag samples, turn off the rear power switch and carefully unplug the power cord.

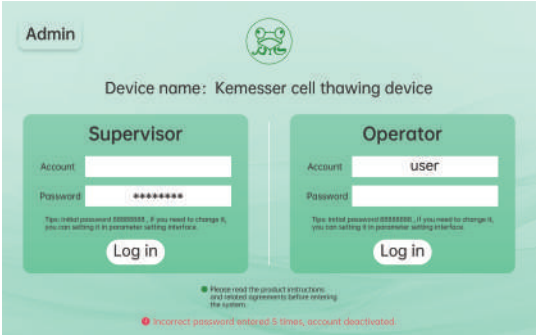
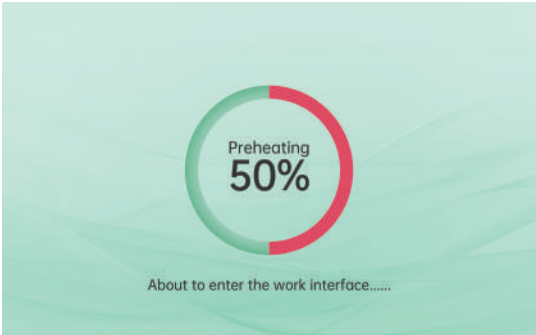
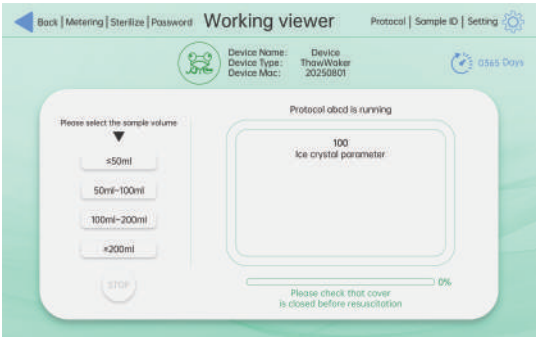


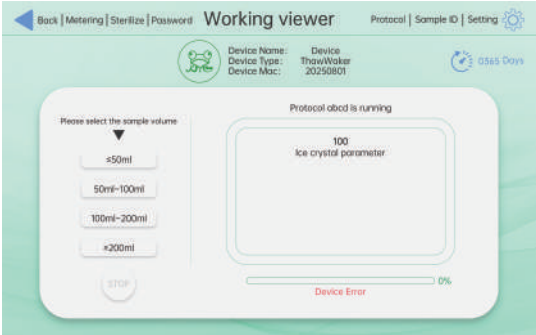
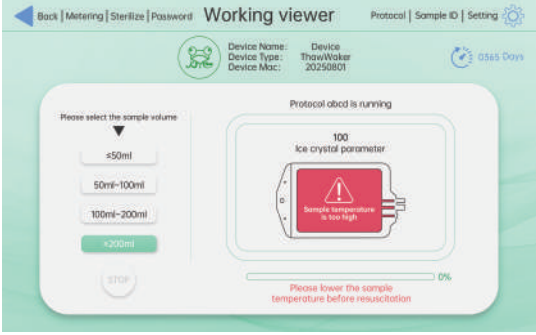
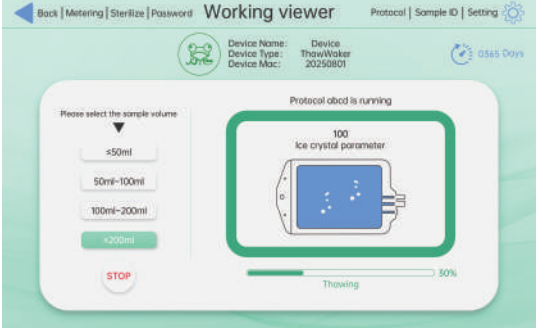
Wipe the device surface, return the device to its place, and keep the surrounding area clean.

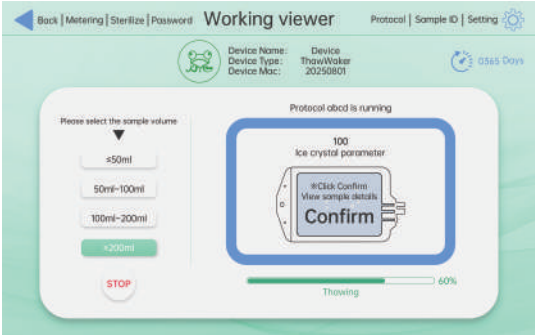
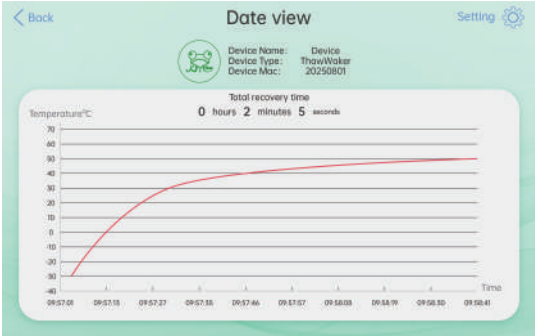
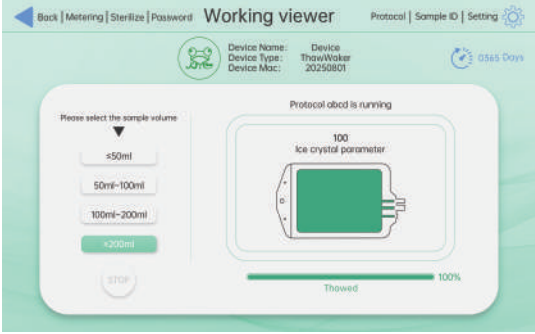
Software Interface

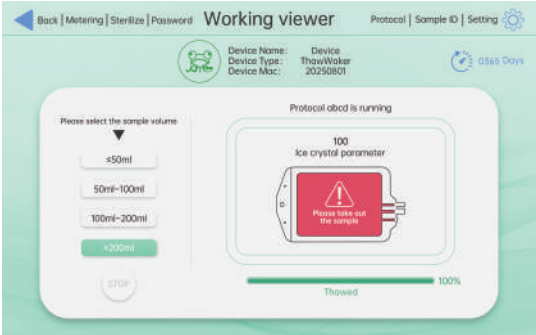
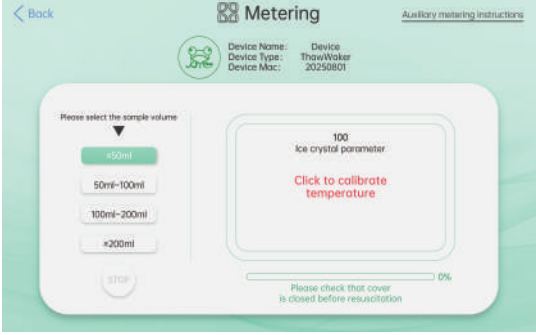
No.	Steps	Display Screen Indication	Sound
1	Check that the instrument is in good condition and connect the power supply;	N/A	N/A
2	Welcome Screen With normal power supply, toggle the switch on the back of the instrument. The display screen will light up automatically, as shown on the right;		N/A
3	Login Screen Click [Admin] at the top left of the login screen to enter the administrator interface.		N/A
4	Admin Interface The device's initial administrator account is Super admin, with the password 88888888.		N/A

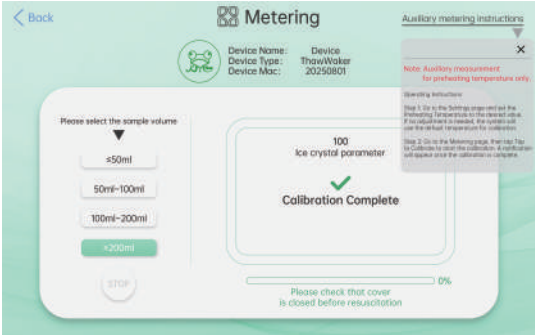
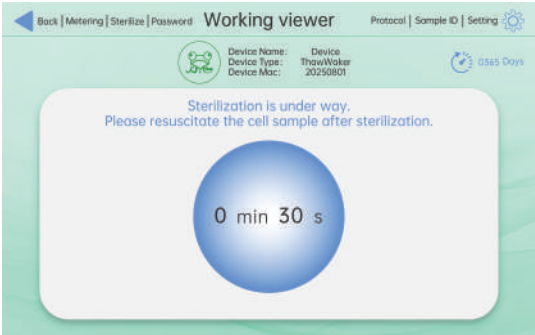
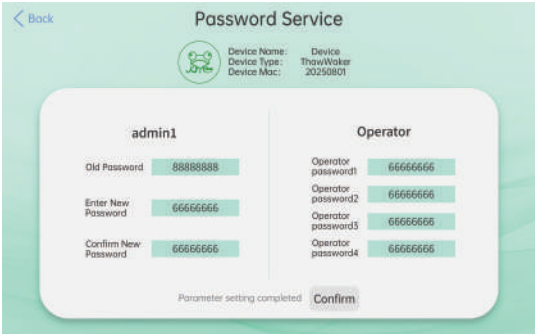
No.	Steps	Display Screen Indication	Sound
5	Incorrect Password If the password is entered incorrectly 5 times, you must wait 60 seconds before trying again.	 The screenshot shows the 'Admin' login interface. At the top, it says 'Admin' with a back arrow. Below that, it displays device information: 'Device Name: ThawWalker', 'Device Type: 20250801', and 'Device Mac: 20250801'. The main login area has fields for 'Account' (filled with 'Super admin') and 'Password'. Below the password field is a 'Log in' button. A red error message at the bottom states: 'Account and password cannot be empty.'	N/A
6	Admin Interface In the administrator interface, you can set the administrator account and password, standby time (if there is no operation within the set time, the device will automatically return to the login screen), password cycle (supervisor and operator passwords have a time limit; if not updated before expiration, the account is locked and the login screen shows a password error requiring higher-level permission to unlock), as well as supervisor and operator accounts and passwords. After modifying this interface, you must enter the admin's old password before clicking Confirm to save the changes.	 The screenshot shows the 'Admin' configuration interface. It has a back arrow and device information. The main area is divided into sections for setting accounts and passwords. It includes fields for 'Admin Account' (Super admin), 'Old Password' (88888888), 'New Password' (66666666), and 'Confirm Password' (66666666). There are also settings for 'Standby Time' (030 min) and 'Password Cycle' (0365 days). Below these are sections for setting supervisor and operator accounts and passwords. A 'Confirm' button is at the bottom right, and a message says 'Parameter setting completed.'	N/A
7	Login Screen Return to the login screen and enter the supervisor account and password.	 The screenshot shows the 'Login Screen' with the title 'Admin'. It displays the device name: 'Kemesser cell thawing device'. There are two login sections: 'Supervisor' and 'Operator'. The 'Supervisor' section has fields for 'Account' (admin) and 'Password', with a 'Log in' button. The 'Operator' section has fields for 'Account' (user) and 'Password', with a 'Log in' button. At the bottom, there is a note: 'Please read the product instructions and related agreements before entering the system.'	N/A

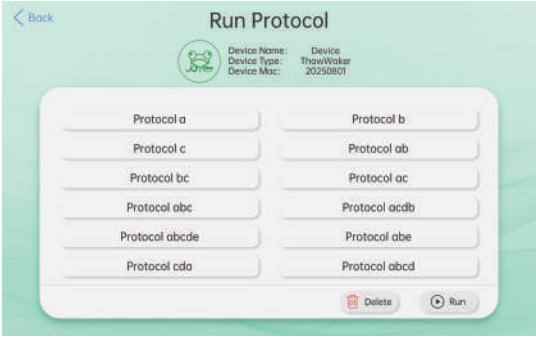
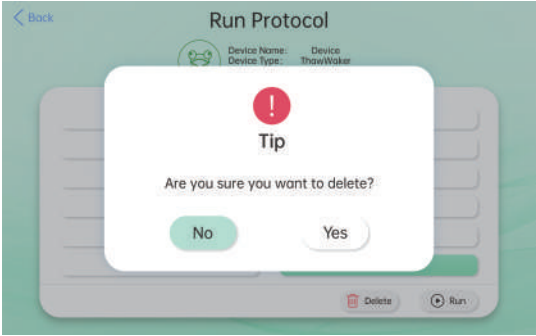
No.	Steps	Display Screen Indication	Sound
8	Incorrect Password If the password is entered incorrectly 5 times, the device will log out the current account (data within the account will be retained).		N/A
9	Preheating Interface After a successful login, the device begins preheating.		N/A
10	Work Interface The work interface includes four sample volume selection buttons, an emergency stop button [STOP], thawing status boxes, thawing progress bars, and more. Ice crystal parameters define the ice crystal size after thawing (the larger the value, the smaller the ice crystals). Click [Back] to return to the login screen to switch accounts. The top right corner displays the remaining days of the password cycle. The running plan currently in progress is shown above the thawing boxes.		N/A

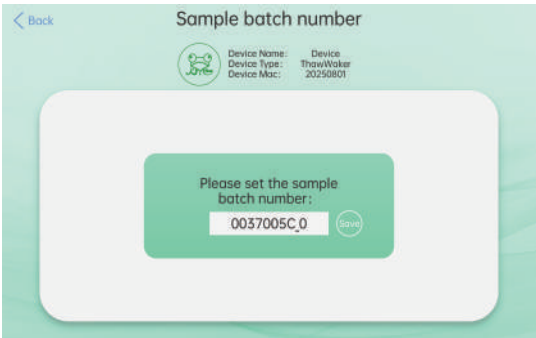
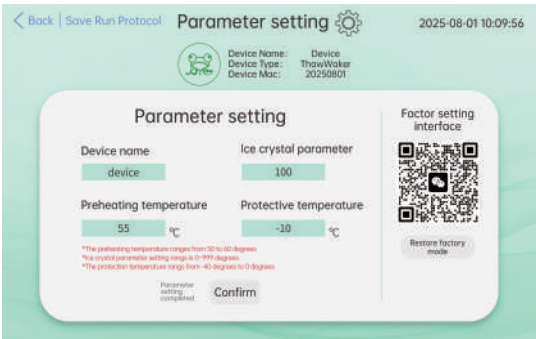
No.	Steps	Display Screen Indication	Sound
11	Work Interface Select the corresponding sample volume button based on the cryobag being thawed. If the device malfunctions, Device Error will be displayed below the thawing progress bar.		N/A
12	Work Interface Place the cryobag into the device and close the top lid. If Sample temperature is too high is displayed, reduce the sample temperature below the set protection temperature.		N/A
13	Work Interface Place the low-temperature cryobag into the device and close the top lid. The device will start thawing, and the interface will flash green, blue, and red alternately.		N/A

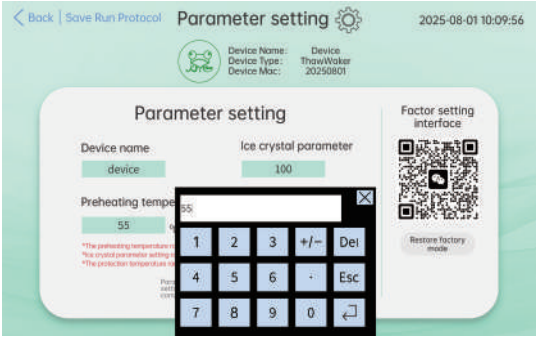
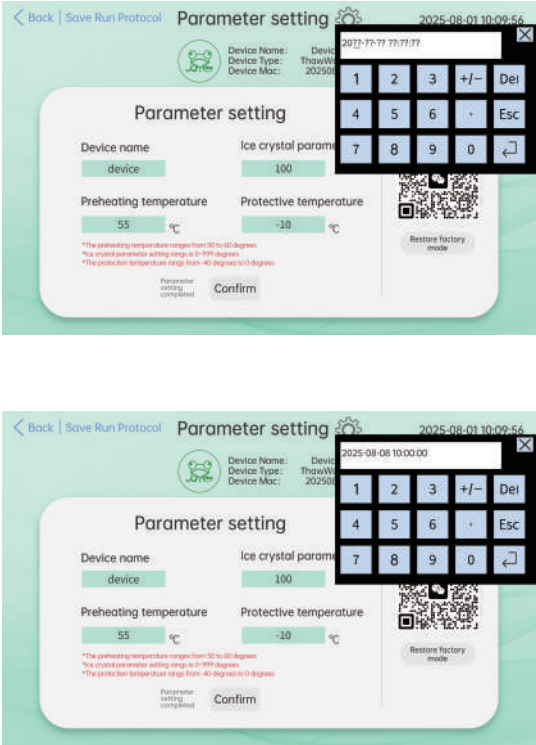
No.	Steps	Display Screen Indication	Sound
14	View Sample Data Click the cryobag icon on the interface. A View Sample Details button will appear; click Confirm to enter the sample data dashboard.		N/A
15	Data view Dashboard The interface displays detailed data curves for the sample, as shown in the image on the right.		N/A
16	Work Interface When the sample thawing is complete, the cryobag icon turns green.		N/A

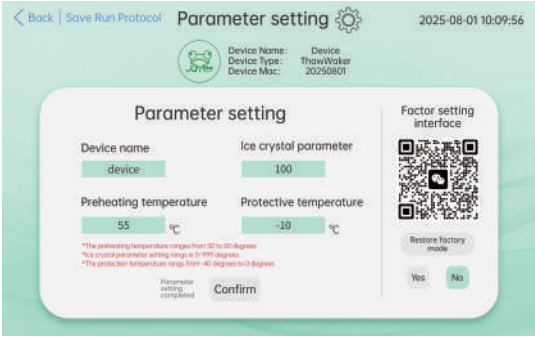
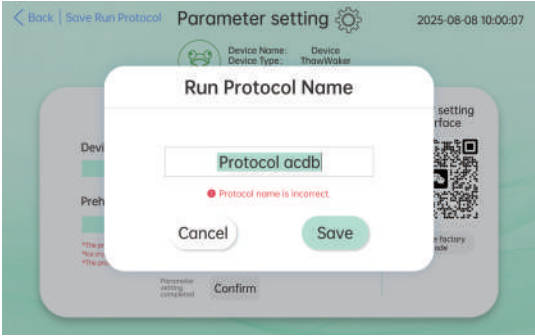
No.	Steps	Display Screen Indication	Sound
17	Work Interface If the cryobag is not removed in time, [Please take out the Sample] will be displayed on the interface.		N/A
18	Measurement Interface Main functions of the measurement interface: • Calibrate the set heating element parameters. • Measurement: After calibration, use a standard instrument to verify the device's accuracy (e.g., if the heating element is set to 55°C, a standard instrument can check whether the actual temperature is correct).		N/A
19	Temperature Calibration Click Calibrate to enter the confirmation interface. After confirming, the calibration process begins. Calibration is complete when the device's heating element reaches the set temperature. (e.g., if the temperature is changed from 45°C to 55°C, enter Service and click Calibrate. The calibration is considered complete only when the heating element stabilizes around 55°C).		N/A

No.	Steps	Display Screen Indication	Sound
20	Metering Click Metering to display the metering instructions.		N/A
21	Sterilization Interface Click [Sterilize] to start the device's ozone function with a 30-second countdown. After sterilization is complete, the interface returns to the main work screen. The interface cannot be operated during sterilization unless the device is powered off and restarted.		N/A
22	Password Service Enter Password Service as a supervisor to set supervisor and operator passwords. After modifying this interface, you must enter the supervisor's old password before clicking Confirm to save the changes.		N/A

No.	Steps	Display Screen Indication	Sound
23	Password Service Enter Password Service as an operator to set the operator password. After modifying this interface, you must enter the operator's old password before clicking Confirm to save the changes.		N/A
24	Run Protocol In the Settings interface, click [Save Run Protocol] at the top left to package the configured parameters into a running plan and save it to this interface. Supervisors can enable or delete running plans, while operators can only enable running plans.		N/A
25	Run Protocol When deleting a run protocol, a confirmation prompt window will appear.		N/A

No.	Steps	Display Screen Indication	Sound
26	Sample Batch Number Enter the batch number and click Save to store the sample batch number.		N/A
27	Settings Interface You can configure: • Device Name • Preheating Temperature • Ice Crystal Parameters • Protection Temperature • Restore Factory Settings You can save run Protocol and set the date and time (long press the date and time at the top right to adjust).		N/A
28	Name and Account Changes • Device series and MAC address cannot be modified. • After completing all settings on the interface, click Confirm to save the changes.		N/A

No.	Steps	Display Screen Indication	Sound
29	Preheating Temperature, Ice Crystal Parameters, Protection Temperature Click the input fields for preheating temperature, ice crystal parameters, or protection temperature in Settings to make changes. After completing all settings on the interface, click Confirm to save the changes.		N/A
30	Date and Time Settings Long press the date and time at the top right in Settings to bring up the date and time input keyboard. Enter the desired date and time, then press Enter to save the changes.		N/A

No.	Steps	Display Screen Indication	Sound
31	Restore Factory Settings In the Settings interface, click [Restore factory mode] at the bottom right. A prompt with [Yes] and [No] will appear, choose according to your needs.	 The screenshot shows the 'Parameter setting' screen with a green header. At the top left are '< Back' and 'Save Run Protocol'. The title is 'Parameter setting' with a gear icon. The top right shows the date and time '2025-08-01 10:09:56'. Below the header, there's a device information section: 'Device Name: Device', 'Device Type: ThawWaker', and 'Device Mac: 20250801'. The main area is divided into two columns. The left column has 'Device name' (set to 'device'), 'Preheating temperature' (set to '55 °C'), and 'Ice crystal parameter' (set to '100'). The right column has 'Protective temperature' (set to '-10 °C') and a 'Factor setting interface' section containing a QR code and a 'Restore factory mode' button. At the bottom right, there are 'Yes' and 'No' buttons for the restore prompt. Small red text at the bottom left provides temperature range warnings.	N/A
32	Save Run Protocol In the Settings interface, click [Save Run Protocol] at the top left to open the input box for the running plan name.	 The screenshot shows the 'Run Protocol Name' dialog box overlaid on the 'Parameter setting' screen. The dialog has a title bar 'Run Protocol Name' and a text input field containing 'Protocol acdb'. Below the input field are 'Cancel' and 'Save' buttons. The background screen shows the same 'Parameter setting' interface as in step 31.	N/A
33	Save Run Protocol If the run protocol name already exists, you need to enter a new name. The interface will display [Protocol name is incorrect].	 The screenshot shows the 'Run Protocol Name' dialog box with the same 'Protocol acdb' in the input field. However, a red error message 'Protocol name is incorrect.' is displayed below the input field. The 'Cancel' and 'Save' buttons are still present. The background screen is the same 'Parameter setting' interface.	N/A

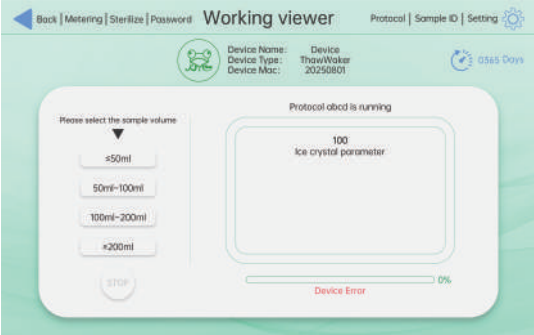
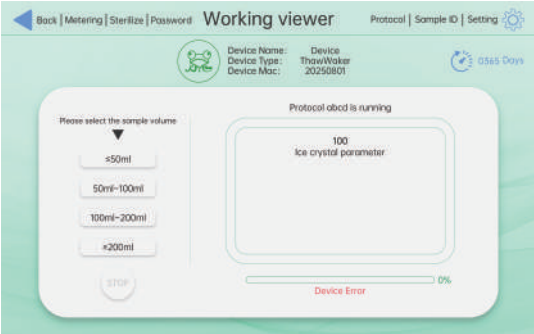
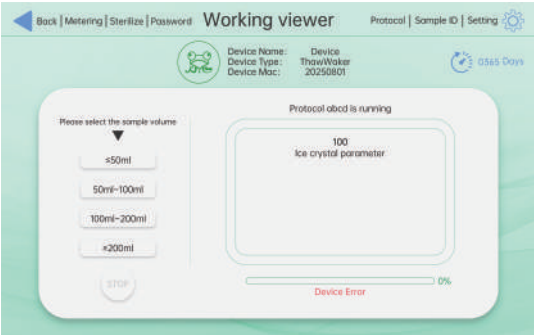
Parameter Settings and Function Usage

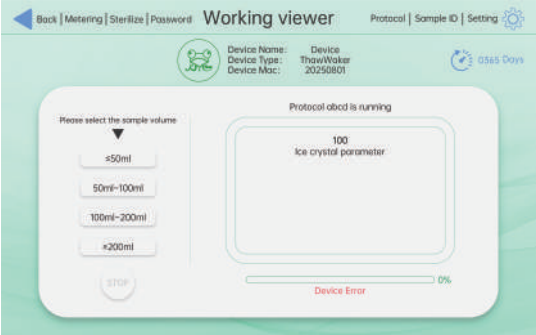
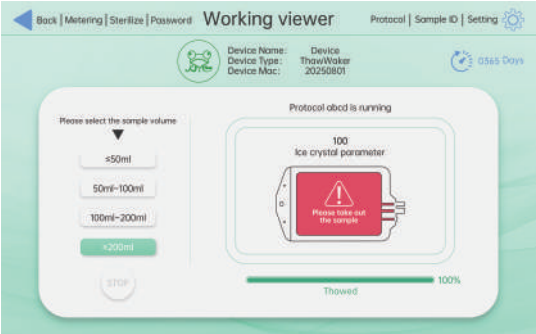
Parameter Settings and Function Usage			
Function Items	Admini	Supervisor	Operator
Assign Supervisor Account	✓	×	×
Assign Operator Account	✓	×	×
Delete (Revoke) Account	✓	×	×
Change Admini Password	✓	×	×
Change Supervisor Password	✓	✓	×
Change Operator Password	✓	✓	✓
Ice Crystal Parameter Settings	×	✓	×
Preheating Temperature Settings	×	✓	×
Protection Temperature Settings	×	✓	×
Modify Device Name	×	✓	×
Restore Factory Settings	×	✓	×
Modify Date & Time	×	✓	×
Measurement Calibration	×	✓	×
One-Click Sterilization	×	✓	✓
Sample Batch No.	×	✓	✓
View Tube Data Dashboard	×	✓	✓
Cell Thawing Operation	×	✓	✓
Bluetooth Printing	×	✓	✓
Enable Run Protocol	×	✓	✓
Save and Delete Run Protocol	×	✓	×
Automatically Save Thawing Data to SD Card	×	✓	✓
Real-Time Temperature Curve Monitoring	×	✓	✓

Note:

1. Administrator can only access the Administrator Interface, cannot perform operations or modify core parameters.
2. Supervisor has full operational permissions, except for the Administrator Interface, which cannot be modified.
3. Operator can only perform Sample Batch No. Setting, One-Click Sterilization, Sample Batch No., Real-Time Temperature Curve Monitoring, Automatically Save Thawing Data to SD Card, and Password Service Interface Settings, cannot modify core parameters.

Alarm List

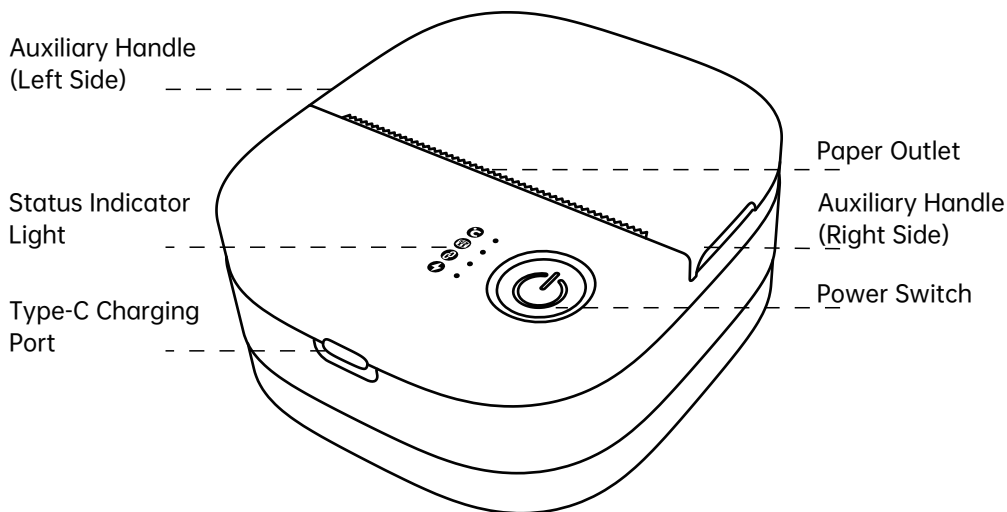
No.	Status and Solutions	Display or Sound Alerts
1	Below the thawing progress bar, "Device Fault" is displayed. The infrared sensor is abnormal. Please Restore Factory Settings. If the error persists after restarting, contact Kemesser technical support.	
2	Below the thawing progress bar, "Device Fault" is displayed. NTC temperature detection is abnormal (temperature reads zero, too high, or too low). Please contact Kemesser technical support.	
3	Below the thawing progress bar, "Device Error" is displayed. The heating element current is too high or zero. Please contact Kemesser technical support to replace the heating unit.	

No.	Status and Solutions	Display or Sound Alerts
4	Below the thawing progress bar, "Device Error" is displayed. The electromagnet current is too high or zero, preventing it from holding the heating unit. Please contact Kemesser technical support.	
5	The interface displays "Please take out sample". If the sample is not removed within 1 minute after thawing is complete, the device will trigger an alarm. The cryobag icon remains green.	
6	The device top lid cannot close because the sample temperature is too high. Remove the high-temperature sample and place a low-temperature sample to try again. If the alarm persists, please contact Kemesser technical support.	The device emits a "beep beep" sound.

Bluetooth Printer Operation Instructions

This device is equipped with a Bluetooth thermal printer for automatically printing cell thawing process reports and preserving critical experimental data.

1. Product Overview



Print Head Temperature

(When the print head temperature is within the normal range, the corresponding indicator remains steadily lit. If the indicator flashes, wait until it turns off and the temperature decreases before continuing use.)



Paper-Out Detection

(The indicator remains steadily lit when paper is loaded in the paper compartment, it flashes when the printer is out of paper.)



Bluetooth Connection

(The indicator remains steadily lit when the Bluetooth connection is successful, it flashes when the connection fails.)



Power Supply

(The indicator remains steadily lit when the device is powered on and operating normally.)

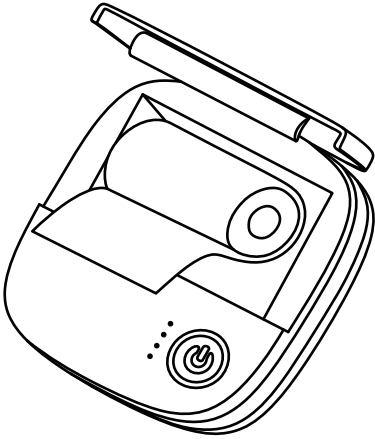
2. Device Preparation

Power Supply: Fully charge the printer or connect it to a USB power source via the Type-C port.

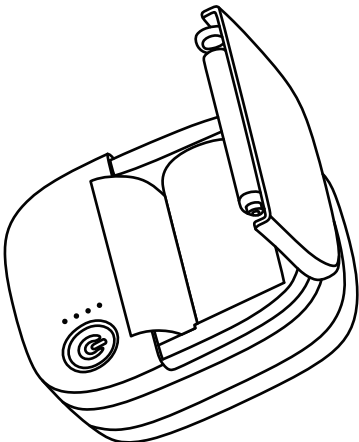
Power On: Press and hold the printer power button to turn on the device. The printer will automatically enter Bluetooth pairing mode.

3. Thermal Paper Installation and Replacement

When the thermal paper is depleted, print quality decreases, or a paper jam occurs, follow the steps below to replace the compatible thermal paper:

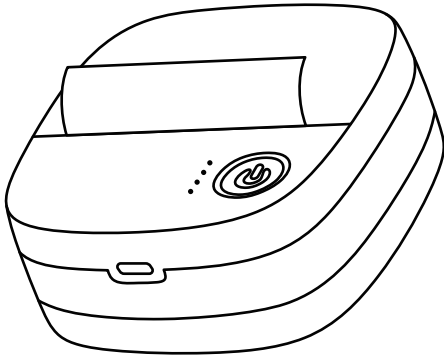


① Open the paper compartment: Pinch the auxiliary handles on both sides of the printer and lift upward to open the top cover.



② Replace the paper roll: Remove the used paper roll and place a compatible thermal paper roll into the paper compartment. Ensure the following requirements are met:

- The smooth printable side of the paper end must face downward.
- The paper feed direction of the roll must face the paper outlet at the front of the device.
- The paper roll must be properly installed without misalignment or jamming.



③ Paper Path Calibration: Pull approximately 2 cm of the paper end outward, then close the top cover. The printer will automatically feed and calibrate the paper. Printing can begin once calibration is completed.

Precautions:

- Please use thermal paper compatible with this device. Low-quality or incompatible paper may cause blurred printing, paper jams, or damage to the print head.
- If the paper is installed in the wrong direction, printing will not function properly and the paper orientation must be adjusted.
- Avoid touching the print head when replacing paper to prevent burns or contamination.

4. Bluetooth Connection and Automatic Printing Process

- ① After the cell thawing instrument is powered on, it will automatically search for and connect to the enabled Bluetooth printer.
- ② When the Bluetooth connection status indicator on the printer (the third indicator light) remains lit, the connection with the cell thawing instrument has been successfully established.
- ③ When the cell thawing process for a single chamber is completed and the sample container is automatically ejected, the printer will automatically start printing and generate a complete report for the thawing session.

5. Print Report Description

The automatically generated thawing report includes the following key experimental parameters:

- StartTime (Thawing start time)
- FinishTime (Thawing completion time)
- ThawTime (Total thawing duration)
- InputTemp (Sample input temperature)
- LeaveTemp (Sample output temperature)
- Port (Chamber number)
- Sample Number (Sample batch number)
- Device (Device name)
- User (Operator)
- MAC (Instrument MAC address)

6. Routine Maintenance and Troubleshooting

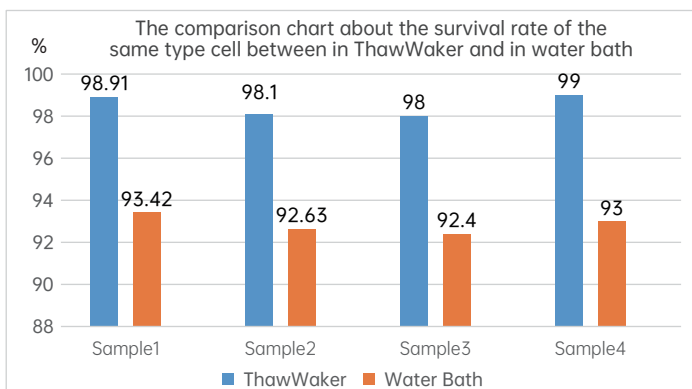
Issue	Troubleshooting and Solution
Bluetooth indicator is off and connection cannot be established	1. Check whether the printer is powered on and has sufficient battery power. 2. Restart the printer and the cell thawing instrument, then reconnect them. 3. Ensure that the printer's Bluetooth connection is not occupied by other devices.
No automatic printing after thawing is completed	1. Check whether the printer is out of paper or the paper is installed in the wrong direction. 2. Clean any dirt or residue from the surface of the print head. 3. Confirm that the Bluetooth connection status is functioning properly.
Printed content is blurred / incomplete	1. Replace with compatible thermal paper. 2. Gently clean the print head with a clean cotton swab to remove dust or stains. 3. Check whether the thermal paper is damp or expired.
Paper jam / abnormal paper feeding	1. Open the paper compartment and remove the jammed paper. 2. Check whether the paper roll is properly installed and recalibrate the paper path. 3. Avoid using damp or wrinkled thermal paper.

Package Contents and Technical Specifications

List	Quantity	Notes
Cell Thawing Device	1	Device
Power Cord	1	220V
SD Card	1	
Card Reader	1	
User Manual	1	

Performance Parameters

Thawing Time	2-8 minutes (varies depending on the cryobag sample)
Rated Power	Maximum 300W
Input/Output Voltage (Power Adapter)	AC 100-240V / DC 12V, 50-60Hz
Output Current	15A
Power Plug	US,UK,EU
Dimensions	L × W × H: 400mmX445mmX255mm
Certification	CE
Thawing Method	Waterless Dry Heat Transfer
Initial Temperature Protection	Yes, users can set it themselves.
Temp. Calibration	Yes, with intelligent calibration for easy third-party measurement.
Sterilize	Yes, the instrument automatically sterilizes from the inside out.
Ice Crystal Control	Customers can adjust the size of ice crystals after resuscitation according to their actual usage needs.
Permission Management	Three Levels: Administrator / Supervisor / Operator
Thawing Complete	With Audible Alert
Display Screen	Touchscreen display; allows real-time monitoring of the thawing chamber.
Data Export	Data is stored on the SD card and can be exported via the host computer.
Online Printing	Equipped with a printer, supports online printing, and can connect directly to the printer.
Permission Levels	Date and time functions are locked, only supervisors can modify them.



Safety Instructions

The automatic cell thawing system is designed for use by trained and experienced laboratory personnel.

1. Read the user manual before installing, using, or cleaning the device.
2. To prevent serious or fatal electric shock, use only the AC power supply provided with this package.
3. Do not place near water or other liquids, and do not immerse the device in water or any other liquid.
4. Unauthorized disassembly of the casing will void the warranty. Do not open the device casing unless authorized by Kemesser.
5. There is a port at the back of the device. Do not insert any plugs into this port or attempt to connect the device to other equipment (e.g., DC power supplies, computers) unless authorized by Kemesser.
6. This device is designed for indoor benchtop laboratory use. Ensure it is securely placed on a level surface. Do not use near strong vibrations, such as centrifuges, shakers, or vortex mixers.
7. Ensure cryobags are frozen on a level surface, with the contents at the bottom of the bag.
8. Ensure low-temperature cryobags are clean and tightly sealed to prevent leakage.
9. Place the low-temperature cryobag into the device slot and start the thawing process. Do not touch the cryobag with hands or any objects until the thawing process is complete.
10. Cryobag thawing generally takes 2–8 minutes. Laboratory personnel are advised to stay near the device to immediately transfer samples once thawing is complete.
11. When not in use, keep the device lid closed.
12. When the device is on, the heating element in the slot is active. Never place any body parts (e.g., fingers) or objects other than the low-temperature cryobag into the thawing slot.

Care and Cleaning

Do not attempt to clean the internal components. Do not insert any cleaning objects or solutions into the open thawing block on the top of the machine. Before or after each use, wipe the exterior with a soft, non-abrasive lab wipe moistened with 70% alcohol. Thawing condensation does not cause problems as it evaporates during the defrosting process or during standby.

Technical Service and Repair

Contact Kemesser technical support at 0571-87156759 or email andy@kemesser.com for help.

Certificate of Conformity:

Certificate of Conformity	
Name:	Automated cell thawing instrument
Model:	
Date:	
Test Result:	Qualified
Inspection date:	

Warranty Card:

Warranty:

1.Enjoy ONE year free warranty for the whole machine from the date of purchase;

2.The following situations are not covered by the warranty:

A: Failure and damage caused by not operating correctly according to the manual;

B: Failures and damages caused by self-disassembly and repair;

C: Failures and damages caused by human factors;

D: If the warranty card cannot be presented or the instrument does not have the official sales seal.

Warranty				
No.	Fault cause	Repair dare	Repair man	Remark
1				
2				
3				
4				
5				
6				
7				



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